

Sophia Jakel

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EDUCATION

Columbia University, School of Engineering and Applied Science *New York, NY – Sept 2024 – May 2028*

Bachelor of Science – Mechanical Engineering, Entrepreneurship Minor, GPA: 3.54

Relevant Courses: Mechanics of Solids, Thermodynamics, Fluid Dynamics, Protection of Industry and IP

Awards: Dean's List

SKILLS

Design: SolidWorks, Fusion360, AutoCAD, FEA

Computer: Python (using NumPy and Pandas), R, LaTeX, Excel, Google Suite, Microsoft Office, Windows and MacOS

Languages: English (native), Russian (native), Italian (fluent), Spanish (conversational)

Manufacturing: Use of 3D printers, laser cutters

RELEVANT EXPERIENCE

Student Venture Associate at Columbia Tech Ventures

New York, NY – June 2026 – Present

- Partner with startups and Columbia research teams to evaluate university-developed technologies and support venture formation and commercialization.
- Track KPIs and portfolio data to monitor commercialization progress; conduct market and competitive research to support early-stage company growth.
- Coordinate entrepreneurship programming and marketing for the 2026 Columbia Climate Summit, connecting founders, researchers, investors, and industry partners.

Columbia University Plasma Physics Laboratory

New York, NY – Sept 2025 – Present

- Engineered and tested cryogenic pellet launchers for fusion energy systems, evaluating hardware design limits and de-risking critical fueling components for next-gen energy devices.
- Support hardware design and vacuum system operation, ensuring optimal performance and reliability.
- Managed technical collaborations with Oak Ridge and Savannah River National Labs to deploy advanced impurity diagnostics, enhancing data fidelity and device outcomes.

Electrical Design Intern at GE Vernova

Clearwater, FL – May 2026 – August 2026

- Redesigned and supported the manufacturing transition of a high-margin instrument transformer line, updating AutoCAD documentation and BOMs with floor operators and quality engineers; conducted industry-standard validation testing and analyzed results to verify safety, performance, and design compliance.
- Spearheaded digitization of GE Vernova's catalogue numbering system into a centralized database, cutting part search time and streamlining part-number creation.
- Selected to be part of the Social Event Planning Committee, organizing engagement and professional development programming for 450+ interns nationwide; partnered with ERGs to secure funding and sponsorship.

Quality Control Engineer at Unified Patents

Washington DC – May – Aug 2025

- Conducted technical due diligence and IP risk assessments on early-stage technology patents; drafted comprehensive patent validity charts to protect clients against IP litigation and validate technology claims.
- Evaluated emerging technology sectors and performed data-driven assessments to guide software tools designed to optimize IP quality and search accuracy.

ACTIVITIES

NCAA Columbia Fencing Team and USA Fencing National Team Member

Sept 2024 – Present

- Member of Columbia University Fencing Team (ranked 2nd in NCAA)
- Compete at the elite international level (2025 Junior World Championships in China), balancing 20+ hours of weekly training with rigorous Columbia SEAS coursework
- Co-founded Columbia Fencing's Social Media Committee, leading team content creation, meet marketing, social media strategy, and promotional campaigns to strengthen team visibility and engagement

Combustion Devices Subteam Member for Columbia Space Initiative

Sept 2024 – Present

- Contribute to the research and design of rocket engine combustion devices using CAD and FEA tools to optimize structural and thermal performance.

Low Voltage Electrician for Columbia Formula SAE (Club)

Sept 2024 – Present

- Collaborate with engineering students and professors to design a wiring harness for a Formula-style race car.
- Utilize industry-standard technology to ensure high performance and reliability in racing conditions.